

Shan CAO

Data scientist with **8+ years** of experience and training from top-tier **business and engineering** schools. Brings **full-stack** expertise across both applications and data.

Paris, France
p2caoshan@163.com
<https://linkedin.com/in/san-cao>
[p2caoshan.github.io/personal_web_san](https://github.com/p2caoshan)

LANGUAGES

English: Professional
French: Intermediate
Chinese: Native

SKILLS & TOOLS

Python, SQL, Spark
Microsoft Azure
ML, LLM, Statistics

HOBBIES

Street Dance
Painting

PROFESSIONAL EXPERIENCE

2026	Coty	Strategy, CMI & Analytics
◆ Build End-to-End Agent flow from scratch: designing DataMart, automating existing data consolidation jobs, (CI/CD) generating regular reports, build agent and MCP to call different tool/skills to support strategic decision and marketing insight. ◆ Marketing analysis and price model.		
2022- 2025	MANULIFE (Forbes Top 250)	Senior Data Scientist
◆ Engineered straight through process (STP) engine of hospital claims for claim team, made use of existing LLM open-source models and OpenAI (RAG) to enhance image and text recognition, combined with NLP and classical ML models on Azure environment. ◆ In charge of fraud, waste, and abuse and reasonable-and-customary models. Worked with doctors, joining statistics, optimization, knowledge graphs, fine-tune and deploy models to predict reimbursement amount. ◆ Manage teams, built end-to-end AI solutions for Automation using Agentic AI(fine tune with SFT and/or RL). ◆ Led data migration for 9 projects across 5 markets to Azure, delivered 8 Personalized Recommendation Campaigns with real-time dashboard, driving revenue growth in Asian markets. Co-developed digital KPI matrix across Asia.		
2019- 2022	MANULIFE (Forbes Top 250)	Compliance Data Scientist
◆ Connected agent-misconduct signals across Southeast Asian markets in one knowledge graph, from SQL and Python into Neo4j, then a frontend compliance teams used. ◆ Applied NLP and PCA to extract insights from call transcripts, categorized easy calls to be handled by chatbot and online process and developed a time-series forecasting model for call volume prediction. ◆ Delivered advanced analytics solutions that streamlined processes and supported strategic decision-making across business units.		
2018- 2019	QIXIN TECH COMPANY	AI Lab Researcher
◆ Designed and developed data ecosystems, cleaning, integrating, analyzing client profiles into data marts for retail clients using classic machine learning algorithms. ◆ Developed recommendation systems to predict preferences and purchasing, enabling personalized marketing. ◆ Published the implemented work as first author at IEEE ICAIBD 2019.		

EDUCATION

2025-2026	ESSEC & CENTRALESUPÉLEC	<i>Data Sciences and Business Analytics</i> <i>Master of Science (Business Intelligence, deep learning, reinforcement learning, decision analysis, etc.)</i> Projects: Marketing Analysis, Causal Analysis
2016- 2018	THE UNIVERSITY OF NEW SOUTH WALES	<i>Systems and Control</i> <i>Master of Engineering Science (optimization, etc.)</i>
2012- 2014	SHANDONG UNIVERSITY	<i>Technology & Instrument of Detection & Control / Finance (Dual-Degree)</i> <i>Bachelor of Engineering / Bachelor of Economics (math, statistics, linear algebra, finance, insurance, etc.)</i>